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Beyond Query Answering *Content-based Recommender Systems*



ITIS Lab – Innovative Technologies for Interaction and Services



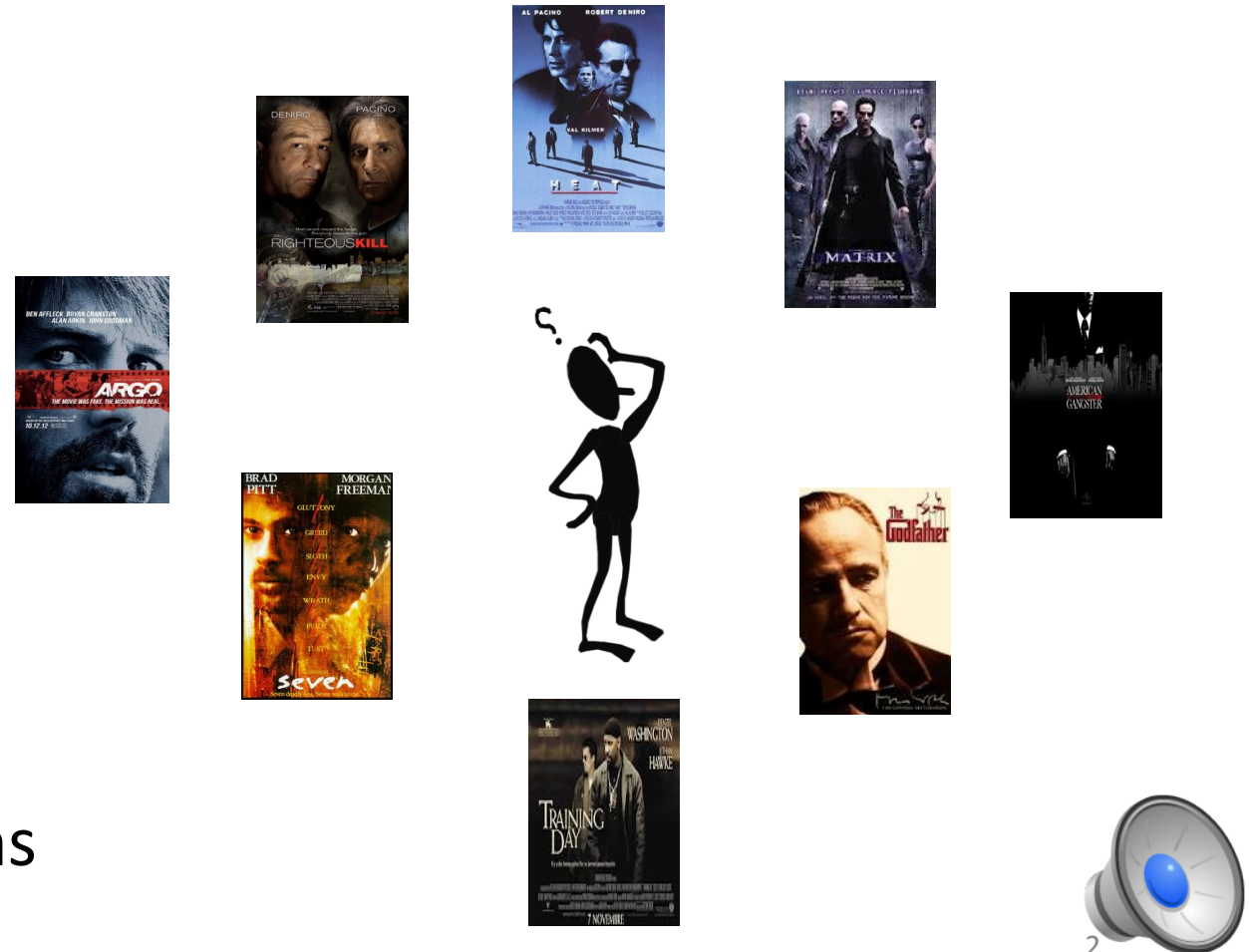
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Università degli Studi di Milano-Bicocca



Recommender Systems

The screenshot shows the Booking.com interface for a search in Berlin. The 'Recommended' filter is selected and highlighted with an orange circle. Below the filter, two hotel listings are shown. The first listing, 'LebensQuelle am Checkpoint Charlie', has a 'Good 7.2' rating and a 'Value Deal' badge. The second listing, 'Hollywood Media Hotel am Kurfürstendamm', has an 'Excellent 8.6' rating and a 'Value Deal' badge. Both listings include details about their location, amenities, and current booking status.

- Help users in dealing with information/choice overload
- Help to match users with items



Examples of Recommender Systems

amazon.it [Iscriviti a Prime](#)

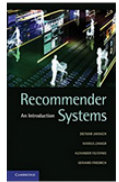
Tutte le categorie ▼

Scegli per categoria ▼

Amazon.it di Vincenzo Offerte Buoni Regalo Vendere Aiuto

Il mio Amazon.it Consigliato in base alla cronologia di navigazione Consigliati per te Migliora i suggerimenti per te

Chi ha acquistato questo articolo ha acquistato anche



Recommender Systems:
An Introduction
Dietmar Jannach
★★★★☆ 2
Copertina rigida
EUR 58,77 ✓Prime

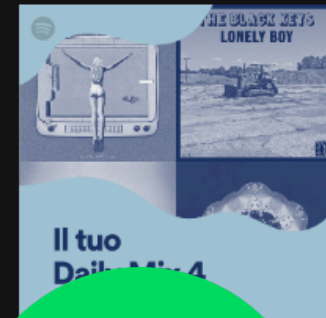
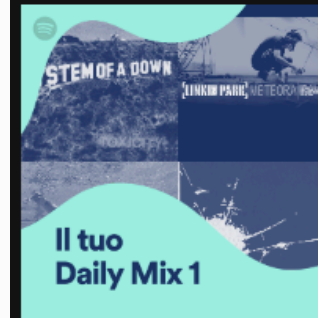


Advanced Analytics with Spark:
Patterns for Learning from Data at...
Sandy Ryza
★★★★☆ 2
Copertina flessibile
EUR 41,60 ✓Prime



I tuoi Daily Mix

Tutta la tua musica che ami, senza il minimo sforzo. Troverai i tuoi beniamini e nuove scoperte.



I Griffin (tema) Canale consigliato per te

[I Griffin Ita - IL BEST SELLER DI BRIAN - Stagione 9 \(Parte 17\)](#)
Lo Strano Tipo
46.752 visualizzazioni • 5 giorni fa

[I Griffin Ita - BRIAN & STEWIE - Stagione 8 \(Parte 17\)](#)
Lo Strano Tipo
65.088 visualizzazioni • 2 settimane fa

[Гриффины - сцены не вошедшие в эфир!](#)
Piter Griffin
307.597 visualizzazioni • 5 giorni fa

[PADRE DE FAMILIA - BUSCADORES DE TESOROS ...](#)
Miralo en Latino
67.447 visualizzazioni • 2 giorni fa

[I Griffin Ita - M COME MALVAGIA MEG - Stagione 8](#)
Lo Strano Tipo
71.432 visualizzazioni • 3 settimane fa

[Гриффины - ЛУЧШИЕ МОМЕНТЫ ч.41](#)
Piter Griffin
58.226 visualizzazioni • 2 giorni fa

Musica rock (tema) Canale consigliato per te

[Queen - Bohemian Rhapsody \(Official Video\)](#)
Queen Official
314.070.590 visualizzazioni • 8 anni fa

[Leonard Cohen - Hallelujah](#)
LeonardCohenVEVO
62.796.670 visualizzazioni • 7 anni fa

[The Cranberries - Zombie](#)
TheCranberriesVEVO
456.510.444 visualizzazioni • 7 anni fa

[Eminem - Lose Yourself \[HD\]](#)
m5vogue23
70.546.979 visualizzazioni • 1 anno fa

[The Weeknd - False Alarm](#)
TheWeekndVEVO
55.386.129 visualizzazioni • 2 mesi fa



COMPUTERS Vs HUMANS

Select music that people can enjoy

Oldest DJ in the world (age 75)



Maximum of 7MIL songs at 95
by superficially listening music 12h/day

DJ Algoriddim (age 1)

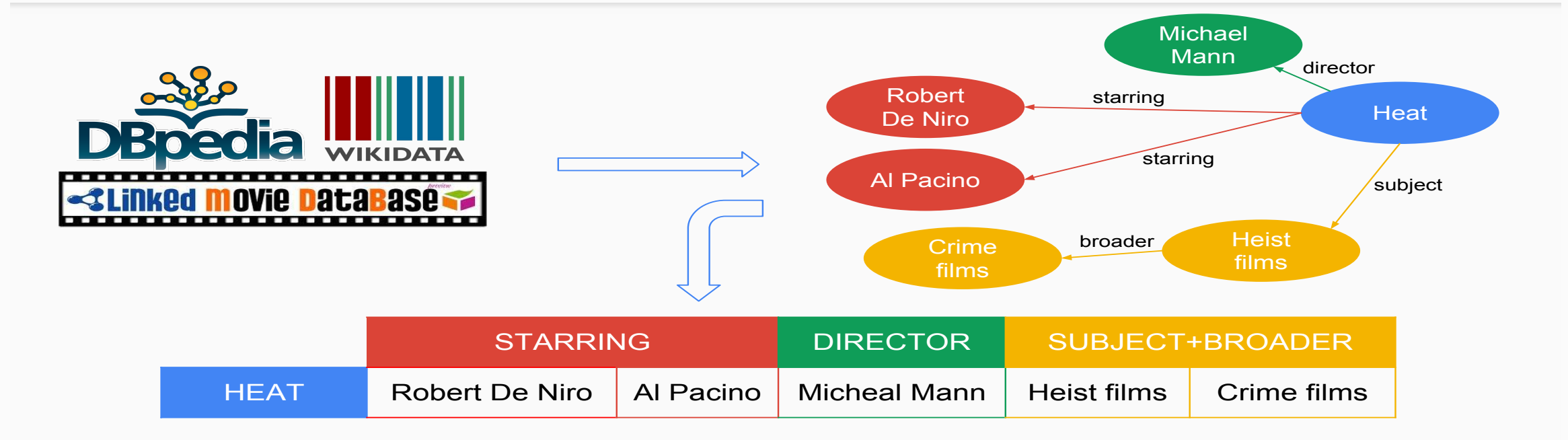


Algoriddim partners with Spotify

Deep analysis of 40MIL songs NOW



Recommender Systems & Semantics



Recommend to a user u items that are similar to items that have been liked by users similar to u

Knowledge Bases may be used to compute similarity between items
(see presentation about similarity)



Similarity

Key point: **evaluate the similarity** between users and/or items

Two approaches:

Collaborative-filtering based: recommend items to the users searching for other users with similar profile (user-based or item-based)

Content-based: recommend items to a user based on the description of the item and the interests of the user (similarity between the recommended item and items that the user has selected in the past)



Why do we need content?

Customers Who Bought This Item Also Bought



Several Recommender Systems
perfectly work **without using any
content!** (e.g. Amazon)

**Collaborative Filtering and Matrix
Factorization** are state of the art
techniques for implementing
Recommender Systems

Recommending New Movies: Even a Few Ratings Are More Valuable Than Metadata

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ABSTRACT

The Netflix Prize (NP) competition gave much attention to collaborative filtering (CF) approaches. Matrix factorization (MF) based CF approaches assign low dimensional feature vectors to users and items. We link CF and content-based filtering (CBF) by finding a linear transformation that transforms user or item descriptions so that they are as close as possible to the feature vectors generated by MF for CF. We propose methods for explicit feedback that are able to handle 140 000 features when feature vectors are very sparse. With movie metadata collected for the NP movies we show that the prediction performance of the methods is comparable to that of CF, and can be used to predict user preferences on new movies.

We also investigate the value of movie metadata compared to movie ratings in regards of predictive power. We compare

1. INTRODUCTION

The goal of recommender systems is to give personalized recommendation on items to users. Typically the recommendation is based on the former and current activity of the users, and metadata about users and items, if available.

There are two basic strategies that can be applied when generating recommendations. Collaborative filtering (CF) methods are based only on the activity of users, while content-based filtering (CBF) methods use only metadata. In this paper we propose hybrid methods, which try to benefit from both information sources.

The two most important families of CF methods are matrix factorization (MF) and neighbor-based approaches. Usually, the goal of MF is to find a low dimensional representation for both users and movies, i.e. each user and movie is associated with a feature vector. Movie metadata (which

(ACM RecSys 2009,
by Netflix Challenge winners)



Content **can tackle some issues** of collaborative filtering

Collaborative filtering - User based

	The Matrix	Titanic	I love shopping	Argo	Love Actually	The hangover
Tommaso	5	1	2	4	3	?
Enrico	2	4	5	3	5	2

How much similar Tommaso and Enrico are?

We need to consider only the movies that **both users have evaluated** and evaluate how much correlated their scores are

$$sim(T, E) = \frac{(5 - 3) \cdot (2 - 3.8) + (1 - 3) \cdot (4 - 3.8) + \dots + (3 - 3) \cdot (5 - 3.8)}{\sqrt{(5 - 3)^2 + (1 - 3)^2 + \dots + (3 - 3)^2} \cdot \sqrt{(2 - 3.8)^2 + (4 - 3.8)^2 + \dots + (5 - 3.8)^2}}$$

$$sim(T, E) \approx -0.73$$



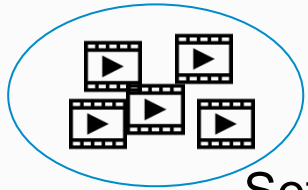
If $sim(T, E) > 0$, T and E are *directly correlated (positively correlated)*

If $sim(T, E) = 0$, T and E are *not correlated*

If $sim(T, E) < 0$, T and E are *inversely correlated (negatively correlated)*

Similarity $\approx$ correlation

Collaborative filtering - User based



Set of all movies

	The Matrix	Titanic	I love shopping	Argo	Love Actually	The hangover
Tommaso	5	1	2	4	3	??
Enrico	2	4	5	3	5	2
Sean	4	3	2	4	1	3
Natasha	3	5	1	5	2	4
Valentina	4	4	5	3	5	2

We look at the **rows** of the score matrix



Correlation
Pearson's corr:
values in [-1,1]

$$\text{sim}(u_i, u_j) = \frac{\sum_{x \in X} (r_{u_i, x} - \bar{r}_{u_i}) \cdot (r_{u_j, x} - \bar{r}_{u_j})}{\sqrt{\sum_{x \in X} (r_{u_i, x} - \bar{r}_{u_i})^2} \cdot \sqrt{\sum_{x \in X} (r_{u_j, x} - \bar{r}_{u_j})^2}}$$

Prediction:

$$\tilde{r}(u_i, x') = \bar{r}_{u_i} + \frac{\sum_{u_j} \text{sim}(u_i, u_j) \cdot (r_{u_j, x'} - \bar{r}_{u_j})}{\sum_{u_j} \text{sim}(u_i, u_j)}$$

Another coefficient:
Jaccard Index

$$\text{sim}(u_i, u_j) = \frac{|R_{u_i}(X) \cap R_{u_j}(X)|}{|R_{u_i}(X) \cup R_{u_j}(X)|}$$

Collaborative filtering - Item based

	The Matrix	Titanic
Tommaso	5	1
Enrico	2	4
Sean	4	3
Natasha	3	5
Valentina	4	4

Tommaso has watched both The Matrix and Titanic. How much similar these movies are?

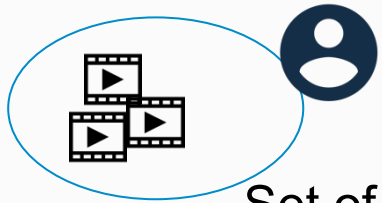
$$\text{sim}(T, E) = \frac{(5 \cdot 1) + (2 \cdot 4) + (4 \cdot 3) + (3 \cdot 5) + (4 \cdot 4)}{\sqrt{5^2 + 2^2 + 4^2 + 3^2 + 4^2} \cdot \sqrt{1^2 + 4^2 + 3^2 + 5^2 + 4^2}}$$

$$\text{sim}(T, E) \approx 0.82$$

Tommaso gave opposite scores to the two movies. However, the similarity between the movies is high, if we consider scores given by all the users



Collaborative filtering - Item based



Set of all movies

	The Matrix	Titanic	I love shopping	Argo	Love Actually	The hangover
Tommaso	5	1	2	4	3	??
Enrico	2	4	5	3	5	2
Sean	4	3	2	4	1	3
Natasha	3	5	1	5	2	4
Valentina	4	4	5	3	5	2

We look at the **columns** of the score matrix

Cosine
Similarity:

$$\text{sim}(\vec{x}_i, \vec{x}_j) = \frac{\vec{x}_i \cdot \vec{x}_j}{|\vec{x}_i| \cdot |\vec{x}_j|} = \frac{\sum_u r_{u,x_i} \cdot r_{u,x_j}}{\sqrt{\sum_u r_{u,x_i}^2} \cdot \sqrt{\sum_u r_{u,x_j}^2}}$$

Prediction:

$$\tilde{r}(u_i, x') = \frac{\sum_{x \in X_{u_i}} \text{sim}(\vec{x}, \vec{x}') \cdot r_{x,u_i}}{\sum_{x \in X_{u_i}} \text{sim}(\vec{x}, \vec{x}')}$$



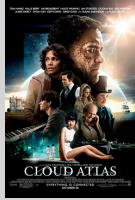
Why do we need content?

					
				✓	
		✓			
			✓		
					✓
	✓				

Collaborative Filtering issues: sparsity



Why do we need content?

					
	✓	✓		✓	
		✓			?
			✓		?
				✓	
	✓	✓			

Collaborative Filtering issues: new item problem



Why do we need content?



Who knows the «customers who bought...»?

Collaborative Filtering issues: poor explanations!



Collaborative filtering - Recap

We use opinions of other users that have common interests

Problems:

- **sparsity** of the score matrix (low probability of finding users who have evaluated the same item)
- **new item / user**
- **gray sheep**: a user with a profile different from the profiles of every other user makes it difficult to find item to recommend to him (no information on similar users)
- **opaque recommendations**





Linked Open Data for Semantics-Aware Recommender Systems

Pierpaolo Basile, Cataldo Musto



Tommaso Di Noia, Paolo Tomeo



tutorial@ESWC 2017 – Portoroz (Slovenia) – 29/05/2017

<https://semanticrecsys.wordpress.com/>



Agenda

Why?

Why do we need **intelligent information access**?
Why do we need **content**?
Why do we need **semantics**?

How?

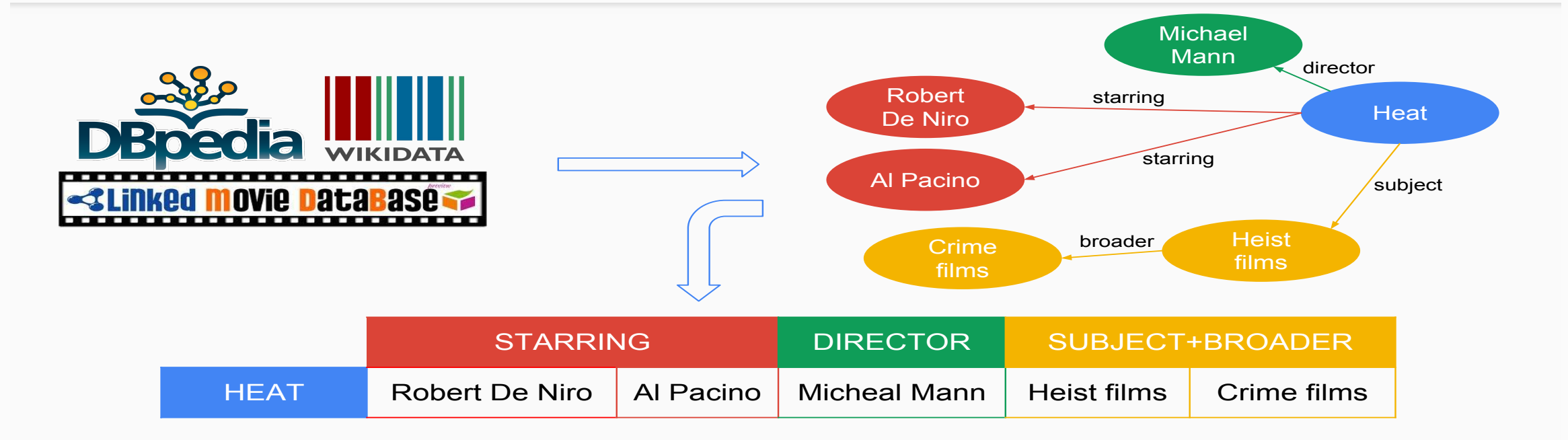
How to **introduce semantics**?
Basics of **Natural Language Processing**
Encoding **exogenous semantics** (top-down approaches)
Encoding **endogenous semantics** (bottom-up approaches)

What?

Recommender Systems based on **Distributional Semantics**
Recommender Systems on **Entity Linking techniques**
Recommender Systems based on **Linked Open Data**
Explaining Recommendations through **Linked Open Data**



Semantic Recommender Systems



Recommend to a user u items that are similar to items that have been liked by users similar to u

KGs may be used to compute similarity between items
(see presentation about similarity)



Content based

Attention on item **descriptions** (descriptive attributes – features)

Different methods to compute the recommendation:

- from Information Retrieval (e.g., cosine similarity)
- from machine learning to learn users' interests from interaction history

Advantages: we can recommend new items, even if not evaluated by any user



Content based - Limitations

Difficult to provide good recommendations if no **discriminant information** is available

Often based on keywords or features to describe objects

What if

- the user selects only items that use few keywords?
- keywords not sufficient to discriminate?
- features are insufficient? (open problem: how to select features?)



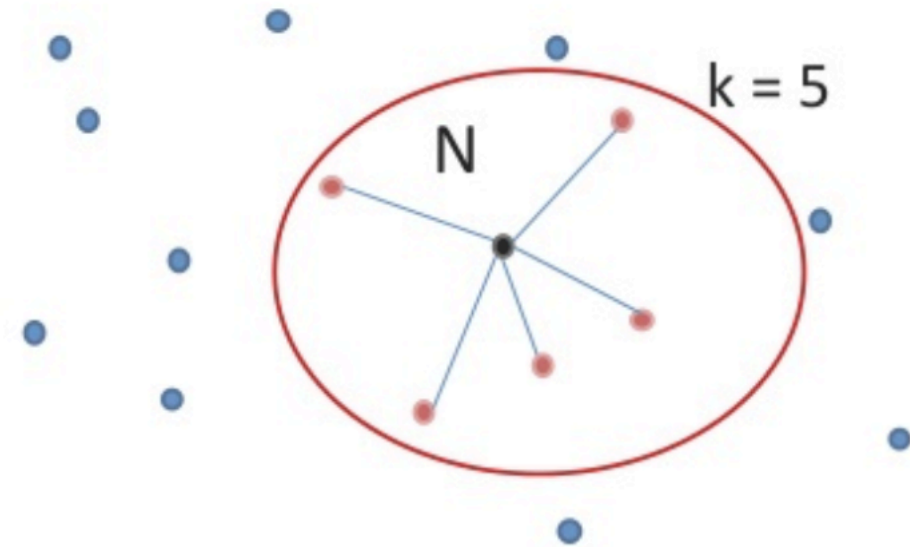
Content-based Semantic Recommendations

- Basic item KNN recommender system
- Given an user u a non rated item i , the rating of i is predicted by:

$$r^*(u, i) = \frac{\sum_{j \in N(i) \cap r(u)} \text{sim}(i, j) \cdot r(u, j)}{\sum_{j \in N(i) \cap r(u)} \text{sim}(i, j)}$$

where:

- $N(i)$ = neighbors of the non rated item i
- $r(u)$ = the items rated by the user u ,
- $r(u, j)$ = the rating value given to the item i by the user u



Similarity functions:

- Jaccard
- Graph kernels
- Cosine similarity in a vector space
- ... several variants

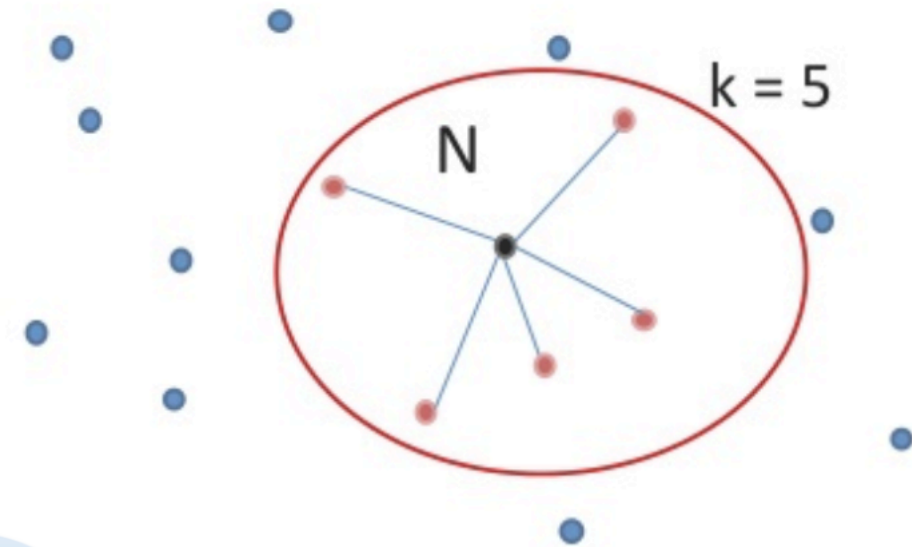
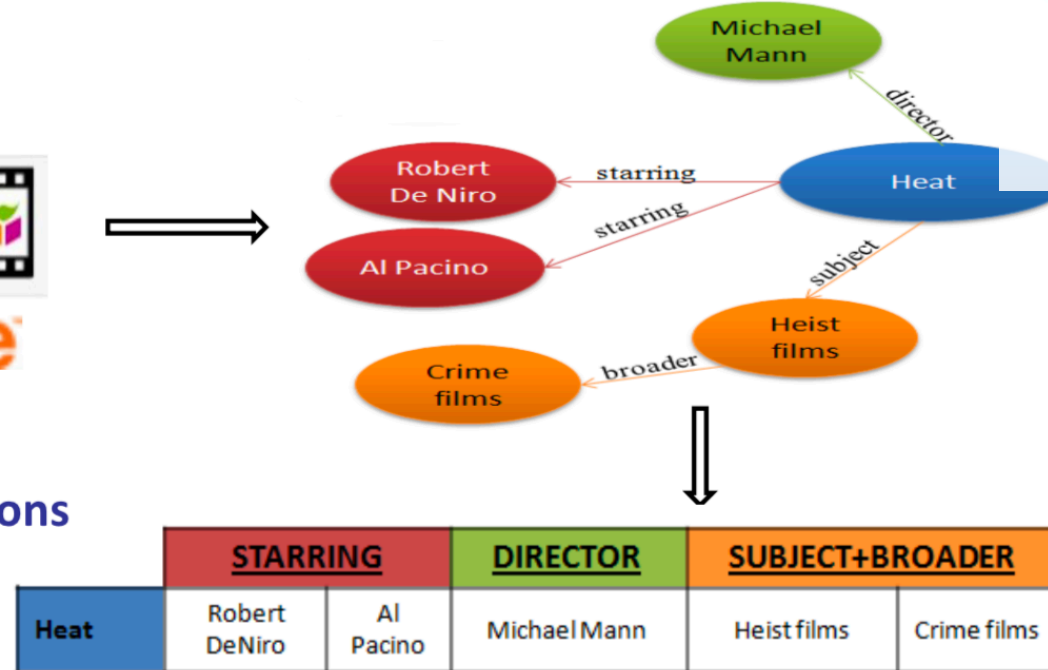


... all based on subgraphs built using certain properties

Content-based Semantic Recommendations



Rich item descriptions



Similarity functions:

- Jaccard
- Cosine similarity in a vector space
- Graph kernels
- ... several variants



... all based on subgraphs built using certain properties

Jaccard Similarity

- Jaccard similarity
 - Measures the values of the selected features that are shared between two items

$$jaccard(i, j) = \frac{|N(i) \cap N(j)|}{|N(i) \cup N(j)|}$$



Which properties should be considered? → Feature Selection Problem

Vector-based Similarity with Weighted Properties

Vector-based model ... (for each property) **cosine similarity** between the property weight vectors of each item

STARRING	Al Pacino (v1)	Robert De Niro (v2)	Brian Dennehy (v3)
Righteous Kill (m1)	✓	✓	✓
Heat (m2)	✓	✓	✗

Righteous Kill (m1)	$w_{v1,x1}$	$w_{v2,x1}$	$w_{v3,x1}$
Heat (m2)	$w_{v1,x2}$	$w_{v2,x2}$	0

$$w_{AlPacino,Heat} = tf_{AlPacino,Heat} \cdot idf_{AlPacino}$$

$$sim_{starring}(\vec{x}_i, \vec{x}_j) = \frac{w_{v1,x_i} * w_{v1,x_j} + w_{v2,x_i} * w_{v2,x_j} + w_{v3,x_i} * w_{v3,x_j}}{\sqrt{w_{v1,x_i}^2 + w_{v2,x_i}^2 + w_{v3,x_i}^2} * \sqrt{w_{v1,x_j}^2 + w_{v2,x_j}^2 + w_{v3,x_j}^2}}$$

$$\alpha_{starring} * sim_{starring}(\vec{x}_i, \vec{x}_j) +$$

$$\alpha_{director} * sim_{director}(\vec{x}_i, \vec{x}_j) +$$

$$\alpha_{subject} * sim_{subject}(\vec{x}_i, \vec{x}_j) +$$

...

=

$$sim(\vec{x}_i, \vec{x}_j)$$

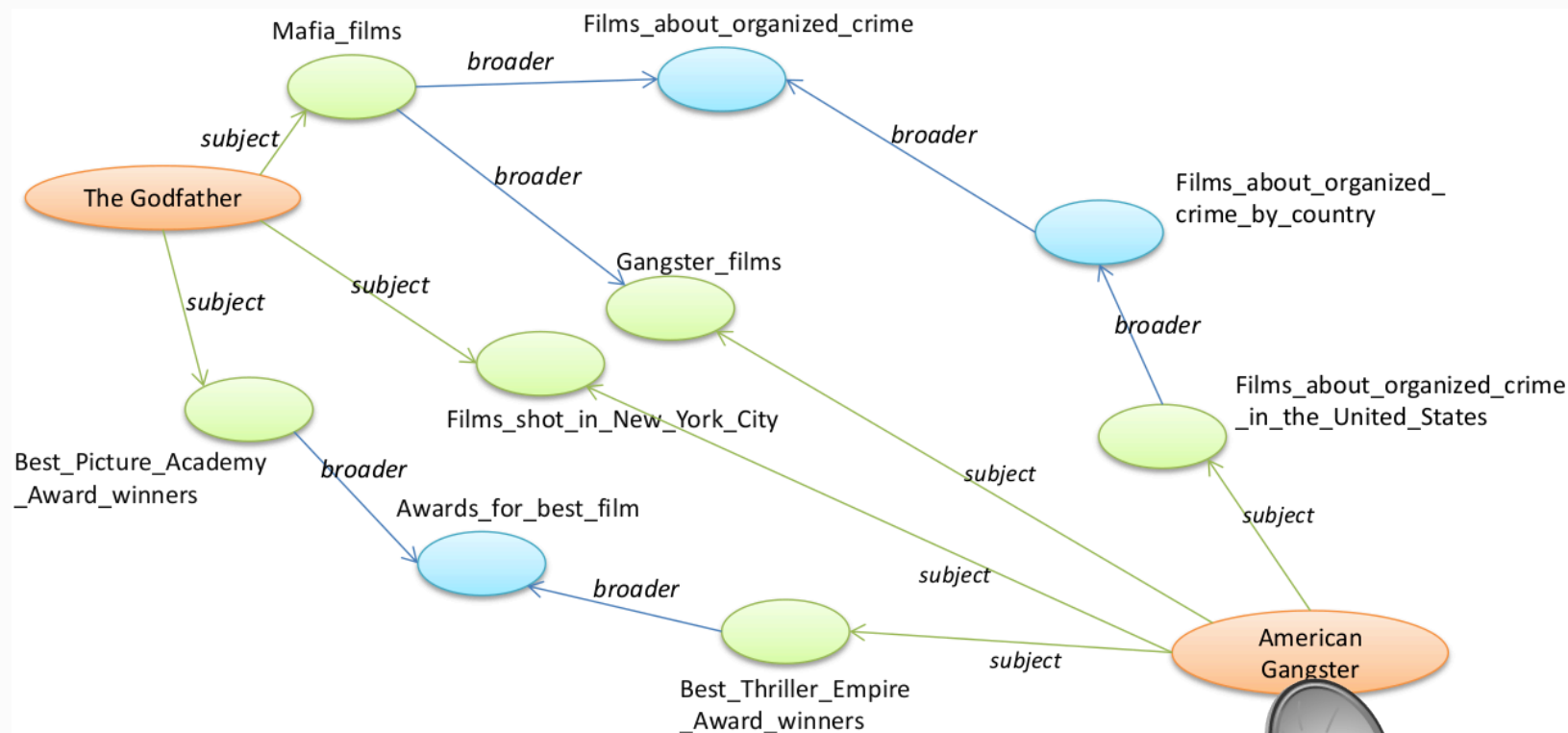
$$tf \in \{0, 1\}$$



Neighborhood-based Representations

Graph-based representation of the entities (h-hop Item Neighborhood Graph)

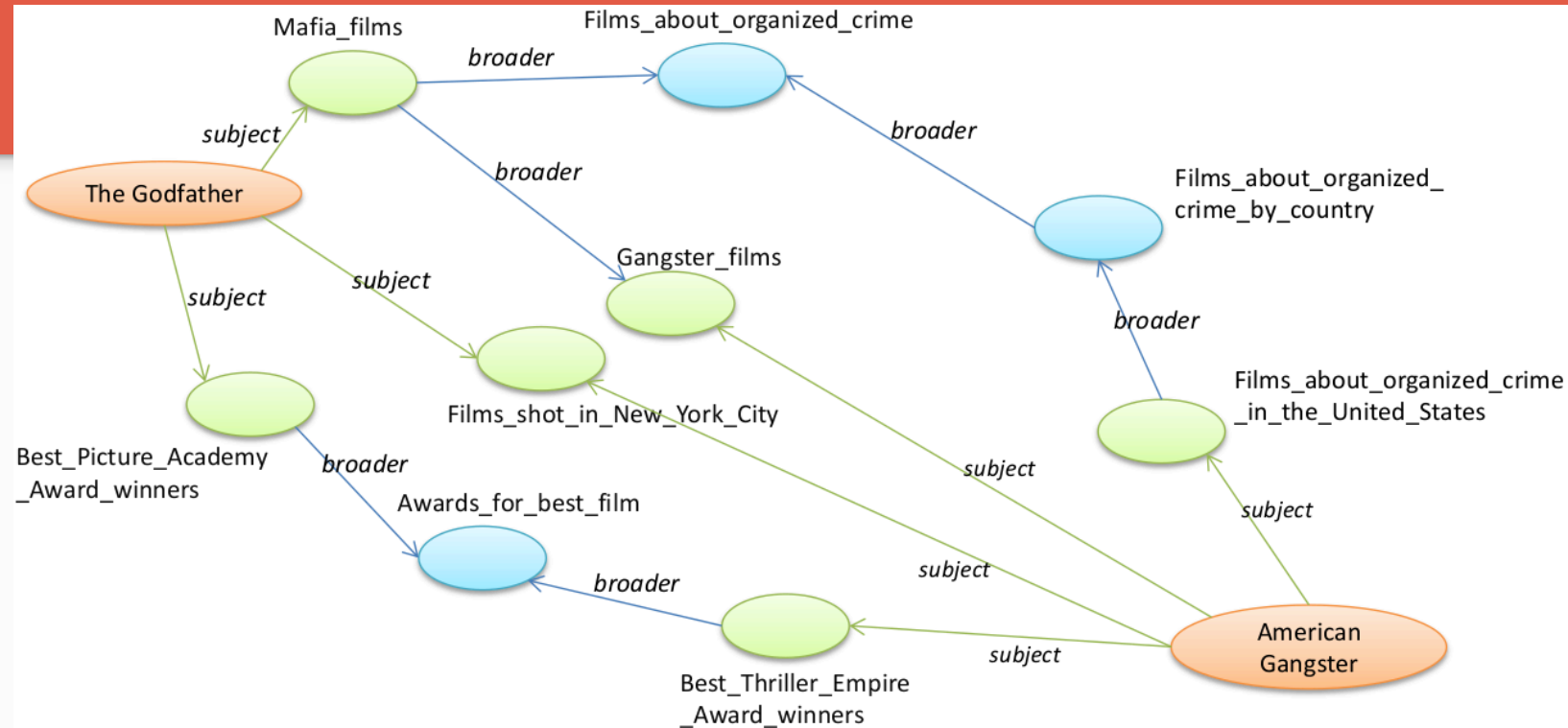
Each entity is identified by itself and by its neighbors



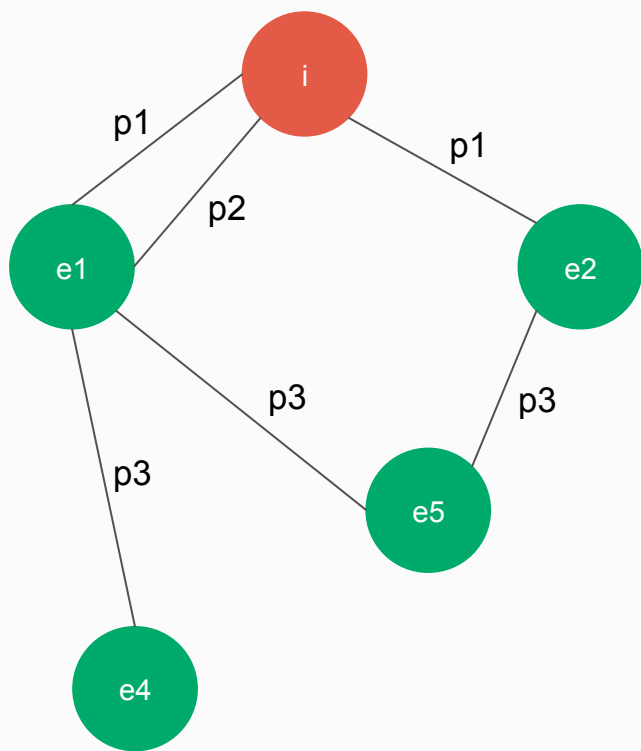
h-Hop Item Neighborhood Graph

For entity i , its neighborhood graph is

- $G^h(i) = (E^h(i), P^h(i))$
- $E^h(i)$ = set of entities reachable in at most h hops from i (shortest path)
- $\hat{E}^h(i)$ = set of entities reachable in exactly h hops from i
- $P^h(i)$ = set of edges reachable in at most h hops from i (shortest path)



h-Hop Item Neighborhood Graph



Generation of sets E and P for $h = 1$ and $h = 2$

$$E^1(i) = \{e_1, e_2\}$$

$$E^2(i) = \{e_1, e_2, e_4, e_5\}$$

$$\hat{E}^1(i) = \{e_1, e_2\}$$

$$\hat{E}^2(i) = \{e_4, e_5\}$$

$$P^1(i) = \{p_1(i, e_1), p_2(i, e_1), p_1(i, e_2)\}$$

$$P^2(i) = \{p_1(i, e_1), p_2(i, e_1), p_1(i, e_2), p_3(e_1, e_4), p_3(e_1, e_5), p_3(e_2, e_5)\}$$

$$\hat{P}^1(i) = \{p_1(i, e_1), p_2(i, e_1), p_1(i, e_2)\}$$

$$\hat{P}^2(i) = \{p_3(e_1, e_4), p_3(e_1, e_5), p_3(e_2, e_5)\}$$



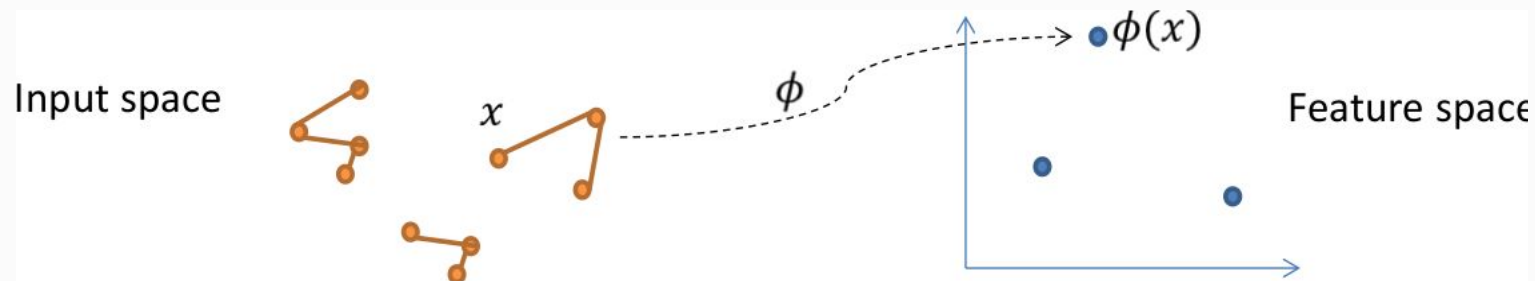
h-Hop Neighborhood-based Graph Kernel

- Towards graph-based similarity

$$k_{G^h}(i, j) = \langle \phi_{G^h}(i), \phi_{G^h}(j) \rangle$$

- This vector product essentially counts entities common in two graphs, weighted by their importance
- We need to map the input in a feature space (so as to support – after normalization – vector-based similarity)
- Each w_{i,e_j} represents the importance of the entity e_j in the neighborhood of the i th entity

$$\phi_{G^h}(i) = (w_{i,e_1}, w_{i,e_2}, \dots, w_{i,e_m}, \dots, w_{i,e_t})$$



Feature Weighting for Similarity Between h-Hop Neighborhood Graphs

- Compute feature e_m for entity i :

$$w_{i,e_m} = \sum_{l=1}^h \alpha_l \cdot c_{\hat{P}^l(i),e_m}$$

- where:

$$c_{\hat{P}^l(i),e_m} = |\{p_k(e_n, e_m) \mid p_k(e_n, e_m) \in \hat{P}^l(i) \wedge e_m \in \hat{E}^l(i)\}|$$

- i.e., **number of edges** that involve the node e_m in the l hop, i.e., occurrences of e_m **in the neighborhood** of distance l
- α is a parameter for **distance-based weighting**: e.g., favor importance of entities closer to i



Computing the Similarity

- Different entity graphs may have different dimensions. If we normalize them, we can obtain cosine similarity [1]
- Otherwise
 - Possible application of algorithms for graph similarity
 - (e.g. Maximal Common Subgraph - MCS [2])



[1] Sergio Oramas, Vito Claudio Ostuni, Tommaso Di Noia, Xavier Serra, and Eugenio Di Sciascio. 2016. Sound and Music Recommendation with Knowledge Graphs. *ACM Trans. Intell. Syst. Technol.* 8, 2, Article 21 (October 2016), 21 pages.

[2] Oramas, Sergio, et al. "A semantic-based approach for artist similarity." *Proceedings of the 16th International Society for Music Information Retrieval (ISMIR) Conference*; International Society for Music Information Retrieval (ISMIR), 2015.

New Approaches for Item-to-item Similarity

RDF2Vec: RDF graph embeddings and their applications

Article type: Research Article

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Abstract: Linked Open Data has been recognized as a valuable source for background information in many data mining and information retrieval tasks. However, most of the existing tools require features in propositional form, i.e., a vector of nominal or numerical features associated with an instance, while Linked Open Data sources are graphs by nature. In this paper, we present RDF2Vec, an approach that uses language modeling approaches for unsupervised feature extraction from sequences of words, and adapts them to RDF graphs. We generate sequences by leveraging local information from graph sub-structures, harvested by Weisfeiler–Lehman Subtree RDF Graph Kernels and graph walks, and learn latent numerical representations of entities in RDF graphs. We evaluate our approach on three different tasks: (i) standard machine learning tasks, (ii) entity and document modeling, and (iii) content-based recommender systems. The evaluation shows that the proposed entity embeddings outperform existing techniques, and that pre-computed feature vector representations of general knowledge graphs such as DBpedia and Wikidata can be easily reused for different tasks.

Keywords: Graph embeddings, linked open data, data mining, document semantic similarity, entity relatedness, recommender systems

Petar Ristoski, Jessica Rosati, Tommaso Di Noia, Renato De Leone, Heiko Paulheim: **RDF2Vec: RDF graph embeddings and their applications.** Semantic Web 10(4): 721-752 (2019)



Evaluate the Recommender System

Information Retrieval measures:

- **precision**: relevant items that have been retrieved over the total number of retrieved items
- **recall**: relevant items that have been retrieved over the total number of relevant items

Ranking quality measures: **nDCG** (for non binary problems)

- Items associated with graded relevance, e.g., [1,2,3] (remark → we can define an ideal ranking)
- Discounted Cumulative Gain (at position p): cumulate gain (gain = rel_i = graded relevance of a retrieved item) and discount this gain as long as the rank position decreases
- normalize DCG by comparing it with Ideal DCG, i.e., DCG of the ideal ranking

$$DCG_p = \sum_{i=1}^p \frac{rel_i}{\log_2(i+1)}$$

$$nDCG_p = \frac{DCG_p}{IDCG_p}$$

Measures from Machine Learning :

- Mean Absolute Error (MAE), Root Mean Square Error (RMSE)



Evaluation: Several Quality Dimensions

Relevance: recommend items that are relevant to the user

Novelty: recommend items that are not known to the user. Often a trade-off between novelty and accuracy

Serendipity: recommend items that are not expected (novelty + surprise). Risk of recommending non relevant items but allows to widen user's interests

Diversity: recommend items that are different increase the likelihood to recommend at least one good item (+ experience)



Evaluation: Several Quality Dimensions

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Serendipity: recommend items that are not expected (novelty + surprise). Risk of recommending non relevant items but allows to widen user's interests

Diversity: recommend items that are different increase the likelihood to recommend at least one good item (+ experience)

Explainability: recommend items that can be explained to (and understood by) the user



KGs for Explainable Recommender Systems

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How to Make Latent Factors Interpretable by Feeding Factorization Machines with Knowledge Graphs

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Vito Walter Anelli , Tommaso Di Noia, Eugenio Di Sciascio, Azzurra Ragone, Joseph Trotta

Conference paper

First Online: 17 October 2019

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Abstract

Model-based approaches to recommendation can recommend items with a very high level of accuracy. Unfortunately, even when the model embeds content-based information, if we move to a latent space we miss references to the actual semantics of recommended items.

Consequently, this makes non-trivial the interpretation of a recommendation process. In this paper, we show how to initialize latent factors in Factorization Machines by using semantic features coming from a knowledge graph in order to train an interpretable model. With our model, semantic features are injected into the learning process to retain the original informativeness of the items available in the dataset. The accuracy and effectiveness of the trained model have been tested using two well-known recommender systems datasets. By relying on the information encoded in the original knowledge graph, we have also evaluated the semantic accuracy and robustness for the knowledge-aware interpretability of the final model.

Vito Walter Anelli, Tommaso Di Noia, Eugenio Di Sciascio, Azzurra Ragone, Joseph Trotta: **How to Make Latent Factors Interpretable by Feeding Factorization Machines with Knowledge Graphs**. ISWC (1) 2019: 38-56

Best Student Paper @ ISWC 2019



Linked open data-based explanations for transparent recommender systems

Cataldo Musto ✉, Fedelucio Narducci 人 ✉, Pasquale Lops ✉, Marco de Gemmis ✉, Giovanni Semeraro ✉

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<https://doi.org/10.1016/j.ijhcs.2018.03.003>

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Highlights

- We design the main components of an *algorithm-agnostic* framework to generate *natural language* explanations;
- We propose a methodology to extract descriptive (direct and indirect) properties about the items, and we use these properties to feed a graph-based explanation model;
- We define a scoring function to rank these explanation patterns and we use the most relevant ones to generate a template-based natural language explanation;
- We validate our methodology by carrying out a large user study (N = 680) in three different domains, as movies, books and music;
- We integrate our methodology in a conversational *recommender system* implemented as a Telegram Bot.

KGs for Explainable Recommender Systems

Cataldo Musto, Fedelucio Narducci, Pasquale Lops, Marco de Gemmis, Giovanni Semeraro: **Linked open data-based explanations for transparent recommender systems**. Int. J. Hum.-Comput. Stud. 121: 93-107 (2019)



Using Ontology-Based Data Summarization to Develop Semantics-Aware Recommender Systems

Authors

[Authors and affiliations](#)

Tommaso Di Noia , Corrado Magarelli, Andrea Maurino, Matteo Palmonari, Anisa Rula

Conference paper

First Online: 03 June 2018

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Properties that optimize content-based similarity

selected with our KG profiling framework
ABSTAT

Abstract

In the current information-centric era, recommender systems are gaining momentum as tools able to assist users in daily decision-making tasks. They may exploit users' past behavior combined with side/contextual information to suggest them new items or pieces of knowledge they might be interested in. Within the recommendation process, Linked Data have been already proposed as a valuable source of information to enhance the predictive power of recommender systems not only in terms of accuracy but also of diversity and novelty of results. In this direction, one of the main open issues in using Linked Data to feed a recommendation engine is related to feature selection: how to select only the most relevant subset of the original Linked Data thus avoiding both useless processing of data and the so called "curse of dimensionality" problem. In this paper, we show how ontology-based (linked) data summarization can drive the selection of properties/features useful to a recommender system. In particular, we compare a fully automated feature selection method based on ontology-based data summaries with more classical ones, and we evaluate the performance of these methods in terms of accuracy and aggregate diversity of a recommender system exploiting the top-k selected features. We set up an experimental testbed relying on datasets related to different knowledge domains. Results show the feasibility of a feature selection process driven by ontology-based data summaries for Linked Data-enabled recommender systems.

Cracking the Feature Selection Problem for KG-based RecSys

Di Noia, T., Magarelli, C., Maurino, A., Palmonari, M., & Rula, A. (2018, June). **Using ontology-based data summarization to develop semantics-aware recommender systems.** In European Semantic Web Conference (pp. 128-144). Springer, Cham.



If interested, check the annexed presentation

Recent Approaches & Deep Learning

- Hybrid approaches: collaborative-filtering + content-based
- Embedding relevant features via Deep Learning
- Several side interesting problems:
 - Check the slides of the Music Recommendation Tutorial on the website

